What is the Solar Potential of Whakaraupō and how can it support resilient communities?

Isla Fisher, Anoushka Paul, Tahliya Nissen-locker, Heidi Stephens, Zachary Taylor

School of Earth and Environment, University of Canterbury.

Lyttelton Energy Transition Society (LETS)

Supervisor: Ann Brower

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Executive summary

The Lyttelton Energy Transition Society (LETS) is currently working towards understanding the solar potential of Lyttelton and assessing the potential of a community focused energy network. The complex geography of Whakaraupō makes the solar potential spatially varied along with socioeconomic factors and current government legislations limiting options for sharing of solar within the community. Mixed method approach was used with GIS, census regressions and case study analysis. The main findings include high solar potential across Lyttelton, Cass Bay, Rāpaki, Governors Bay and Diamond Harbour.

Socioeconomic factors such as detached housing and age of residents influence the patterns of solar installations. A community-based approach could help provide areas of low potential with solar generated power, but current legislations limit the options for this. Recommendations for LETS include focusing on including a community hub with a battery and investigating the Gebbie's Pass grid connection for a possible solar farm.

Introduction

Solar photovoltaic (PV) installations create the opportunity for a reduction in reliance on centralised energy production and strengthening of the energy resilience of communities. Improvement of energy resilience through PV depends on physical solar resources, potential for new instalments, electrical network capacity, and arrangements for storing and sharing.

Whakaraupō houses settlements along the coast including Lyttelton, Cass Bay, Rāpaki, Governors Bay, and Diamond Harbour which are the focus of this study as seen in Figure 1. Figure 2 demonstrates the complex volcanic terrain of Whakaraupō, which leads to spatial variation of solar radiation with contributing factors of aspect, terrain, slope and existing developments (Corripio, 2003). As well as this, socioeconomic conditions widely influence solar use alongside network and policy regulation constraints. LETS sought to understand how solar installations could improve energy resilience within Whakaraupō.

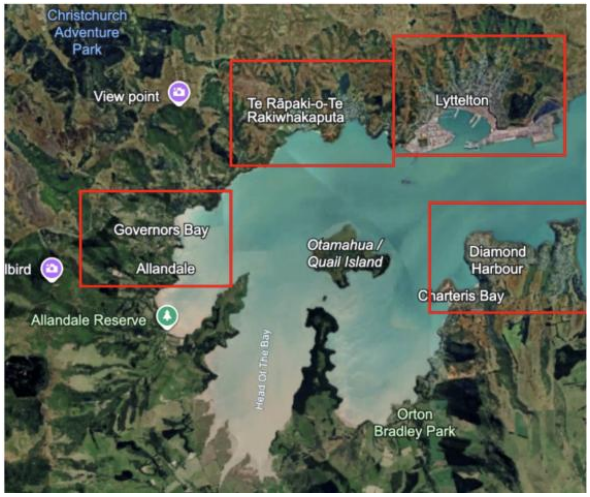


Figure 1: Study area including Lyttelton, Rāpaki, Cass Bay, Governors Bay and Diamond Harbour. Red boxes indicate the clipped area for each study.

This report addresses the question '*What is the Solar Potential of Whakaraupō and how can it support resilient communities?*' by using spatial analysis of solar potential, demographic patterns, community initiatives, and sharing opportunities. ArcGIS Pro was used to address this, allowing for comparisons between current installations and areas of high potential to provide insight into how future installations can increase energy resilience. It also investigates network constraints on future installation and community arrangements for sharing and self-governance. Overall, this report seeks to inform LETS on the solar potential of Whakaraupō and identify priorities for further research.

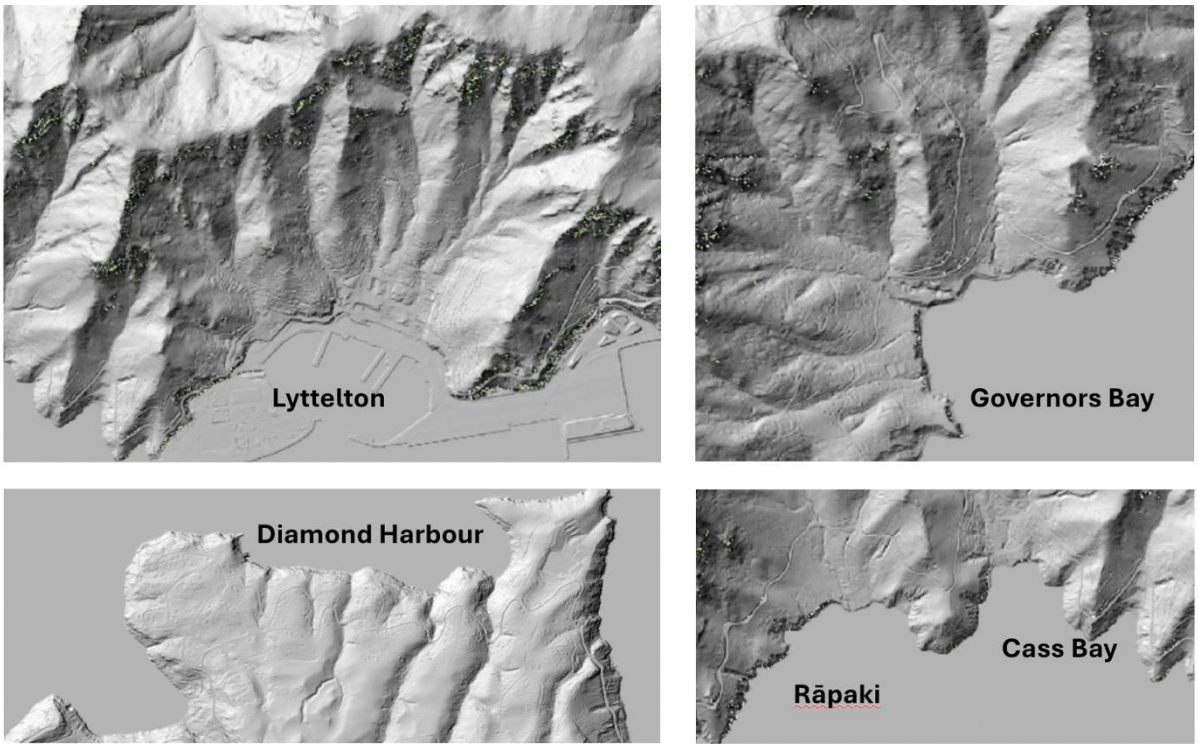


Figure 2: Hill Shade maps of each study area location showing the complex terrain.

Chapter 1 – Where are existing solar installations located

1.1 Literature review

Rooftop solar is greatly influenced by physical factors such as topography, slope, and aspect and climate factors such as cloud cover and atmospheric conditions. Physical factors are particularly important in areas with complex terrain, where surrounding hills can block sunlight.

Corripio (2003) uses slope and terrain to calculate how much solar radiation reaches any given surface and found that areas with complex terrain have the most scattered solar potential. This is relevant to rooftop solar because differences in surrounding terrain can result in substantial spatial variation in solar exposure. Ehnberg and Bollen (2004) highlight that atmospheric conditions are constantly changing, especially in mountainous environments with complex terrain, and use cloud observations to highlight different cloud cover influences the solar received. Solar potential assessments benefit from considering both the physical characteristics and atmospheric conditions affecting incoming solar radiation.

1.2 Methods

Manual detection was used to map the existing solar panels in Whakaraupō using ArcGIS Pro. Chapters 1 and 2 use 0.075m satellite imagery from September 2025, building outlines from May 2025, 1m Digital surface model (DSM) from April 2025, and a 1m Digital elevation model (DEM) from March 2025. The DEM was used to create the Hillshade maps seen in Figure 2.

Create Feature tool was used to manually create polygons of visible rooftop solar panels in the satellite imagery. The solar panel feature class was joined to the building outlines. Then, *Select by Attributes* tool was run creating a new layer where join count of the installed panels was greater than 0, resulting in a category with defining each house with solar panels, rather than just individual installations. This was repeated for the whole study area as seen in Figures 3-6 and quantified in Table A1.

1.3 Results

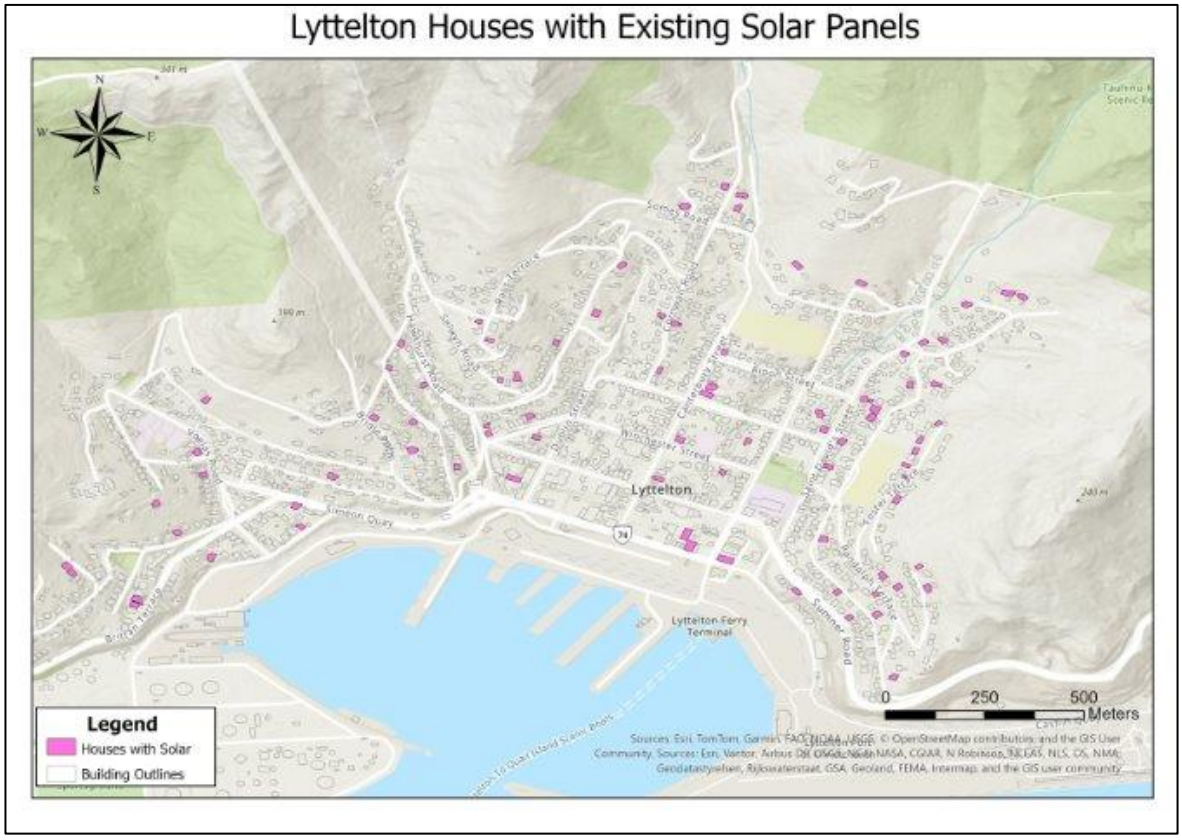


Figure 3: This shows existing solar installations in Lyttelton. There are little pattens with some light clustering on the eastern slopes.

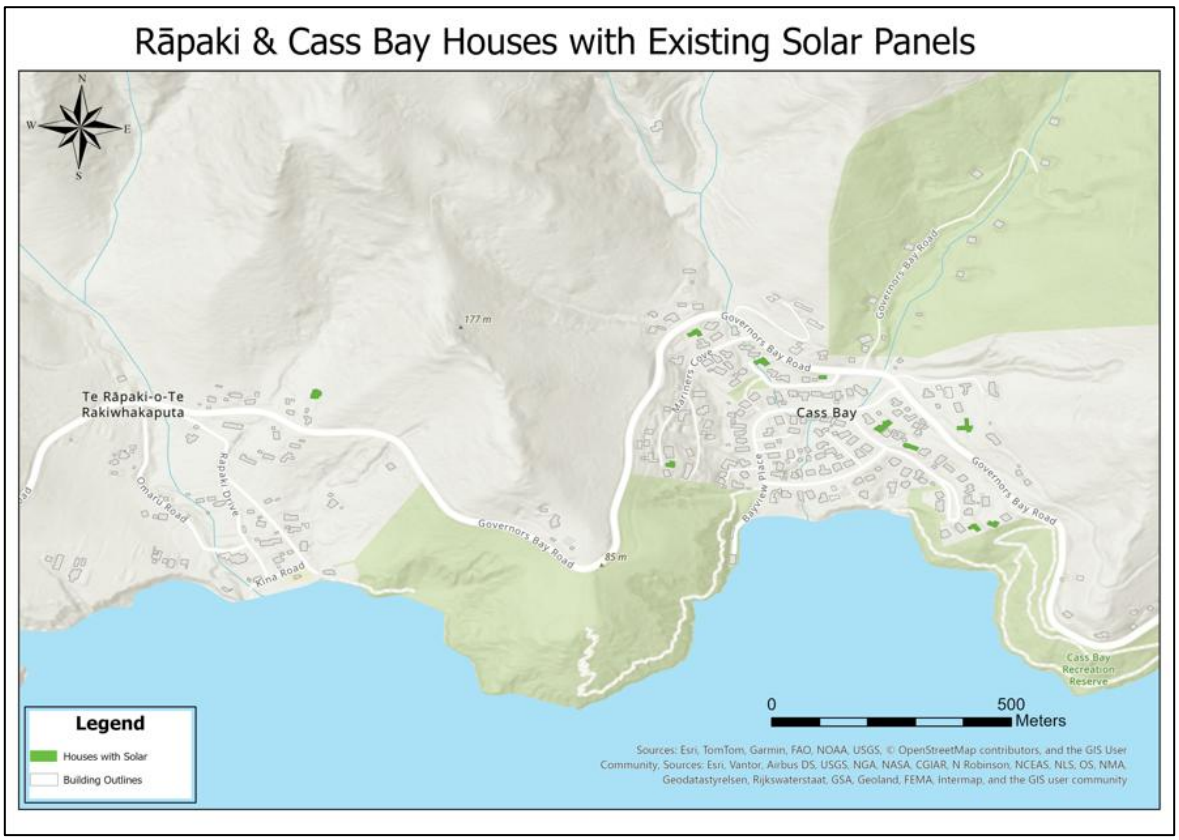


Figure 4: This shows existing solar panels in Rāpaki and Cass Bay. In Rāpaki there is only one house with a solar panel and a few scattered around Cass Bay.

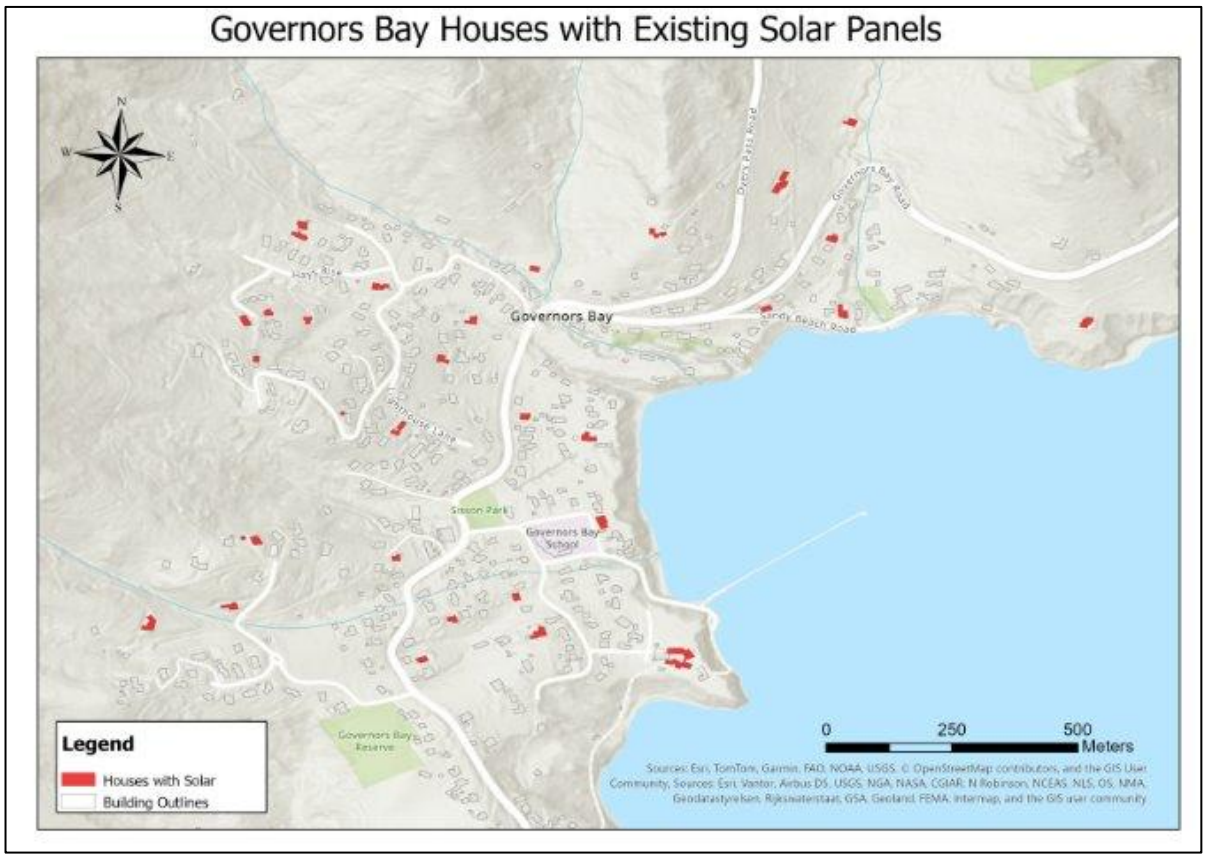


Figure 5: This shows existing solar installations in Governors Bay. There is a dispersed pattern with some panels up on the hill and some on the shoreline.

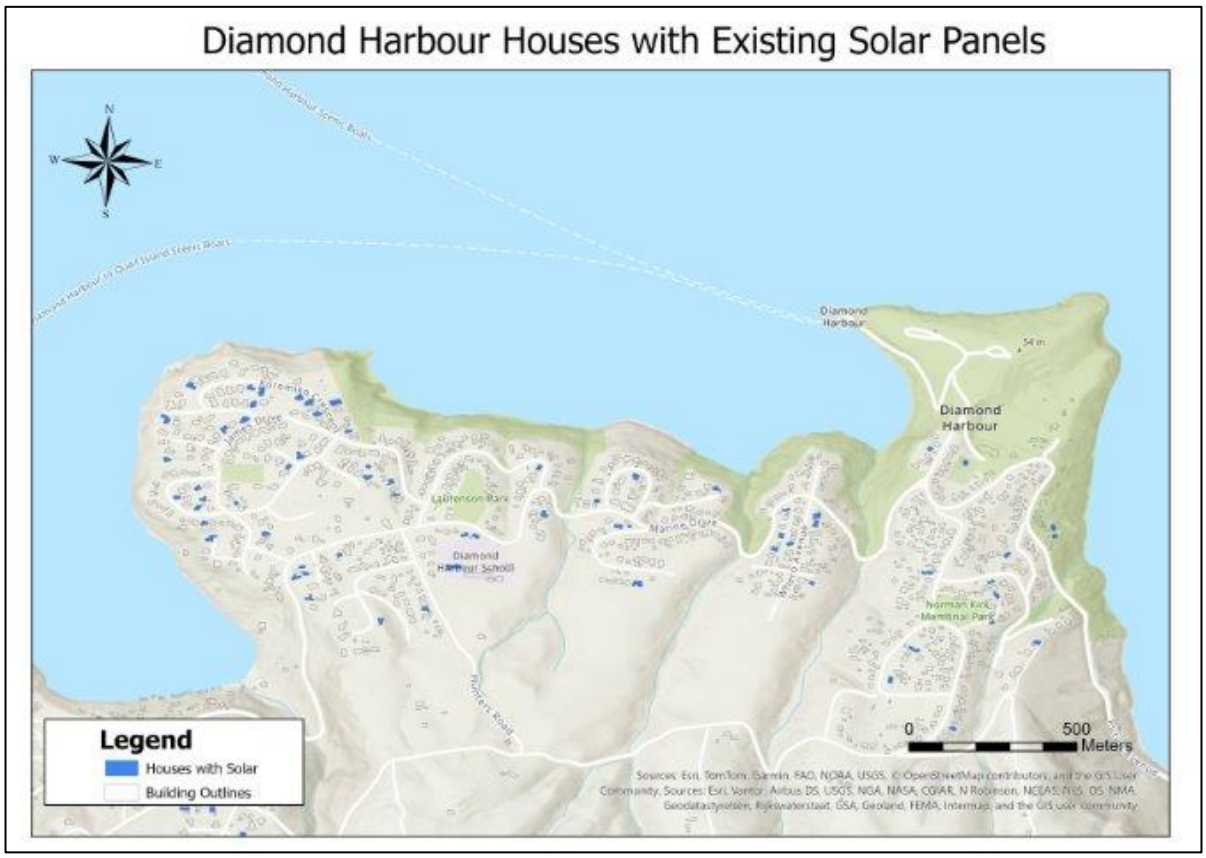


Figure 6: This shows existing solar panels in Diamond Harbour. There is a cluster on the west side of the bay and less, mor dispersed on the east side of the bay.

Chapter 2 – How does physical solar potential vary across Whakaraupō?

2.1 Methods

Feature Solar Radiation and *Raster Solar Radiation* tools on ArcGIS Pro estimated the solar potential of Whakaraupō. Both tools used a LiDAR derived DSM and building outlines as input to calculate the theoretical solar potential of rooftops, the same method used in Margolis et al. (2017) and Suomalainen et al. (2017). A diffuse proportion of 0.3 and transmittivity value of 0.5 were used to reflect "generally clear sky conditions" similarly to Suomalainen et al. (2017). An annual period was chosen between 1st January 2025 and 1st January 2026 to capture seasonal fluctuations. The *Feature Solar Radiation* tool output building polygons with attribute information on total and average solar radiation values. The average radiation was converted to a raster layer using the *Polygon to Raster* tool to produce Figures 7-10.

The *Raster Solar Radiation* tool outputs a raster where each individual pixel had a corresponding radiation value (Esri, n.d.a.). To quantify the theoretical potential, *Zonal Statistics as Table* tool was used with the radiation raster and a building outline buffer layer as input. Building outline buffer was created using the *Buffer* tool with a value of minus one, so the perimeter of the building footprint was decreased by one metre to remove pixels which represented walls of a building due to radial displacement error (Esri, n.d.b.). *Zonal Statistics as Table* tool was used to create the theoretical columns in Table A2.

To calculate a more accurate theoretical potential of Whakaraupō, the raster solar radiation layer was clipped by slope, aspect, and area values, as suggested by Margolis et al. (2017). Slope and Aspect layers were created with corresponding *Spatial Analyst* tools and DSM as input. To filter by area, a binary raster with 0 for 'NoData' and 1 for all other values was input into the *Region Group* tool which grouped the binary raster into regions of continuous patches. In New Zealand, south facing roofs receive less sun than north, east, or west facing roofs (BRANZ, 2007; Hoare, 2020; Current Generation, n.d.). According to Harrisons Solar, the optimal slope for solar panels in Christchurch is 32 degrees in summer and 57 degrees in winter, and for New Zealand between 23-36 in summer and 51-61 in winter (BRANZ, 2007). Solar panels can be mounted, so flat roofs are not excluded (BRANZ, 2007; Hoare, 2020; Kāinga Ora – Homes and Communities, 2022). The smallest rooftop area considered by Suomalainen et al. (2017) was 14m^2, given a common panel area is around 7m^2. *Raster Calculator* tool was used to filter the radiation raster to pixels that had a slope less than 60 degrees, a non-south facing roof (not south, south-east, or south-west), and an area of at least 14m^2. *Zonal Statistics as Table* tool was then used to create the filtered columns in Table A2.

2.2 Results

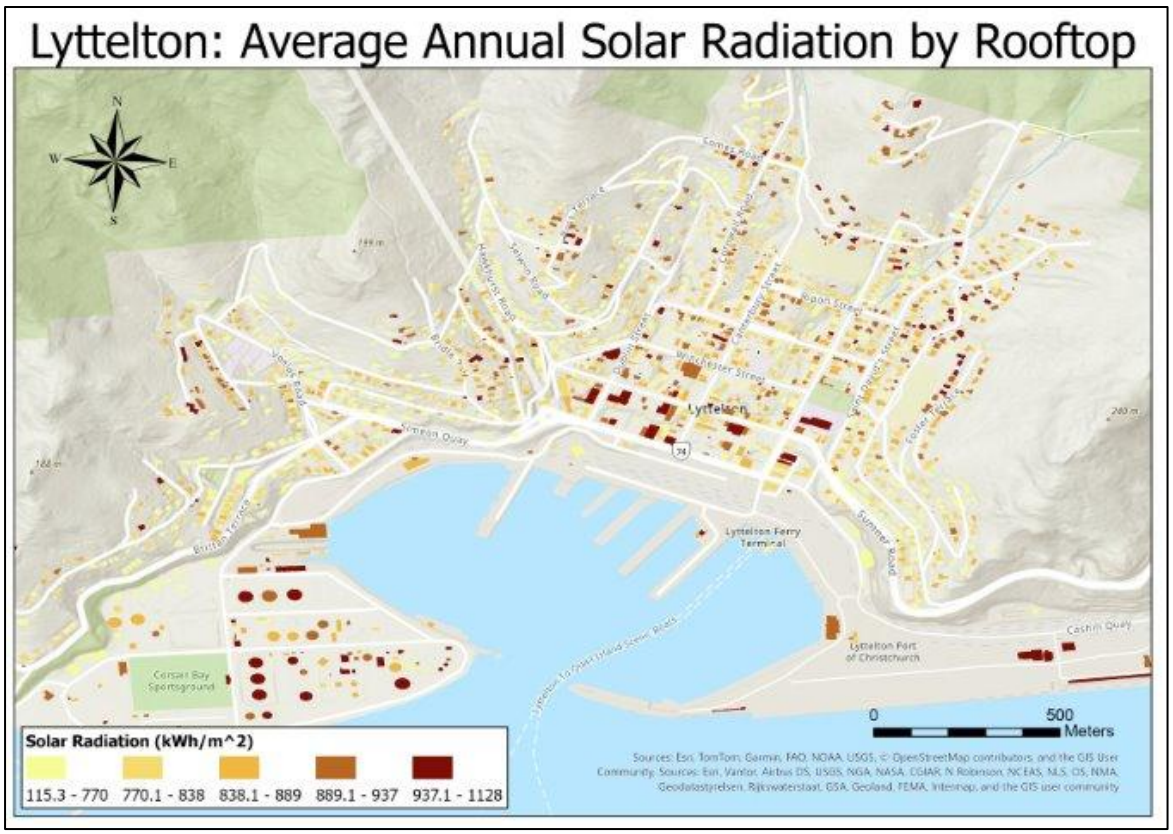


Figure 7: This shows a map of average annual solar radiation by rooftop in Lyttelton. Figures 7-10 were created using ArcGIS Pro Feature Solar Radiation Tool. Each rooftop is coloured to show the average solar radiation received in kWh per m^2. The category ranges are representative of five quantiles where each category includes 20% of buildings (considering the whole study area).

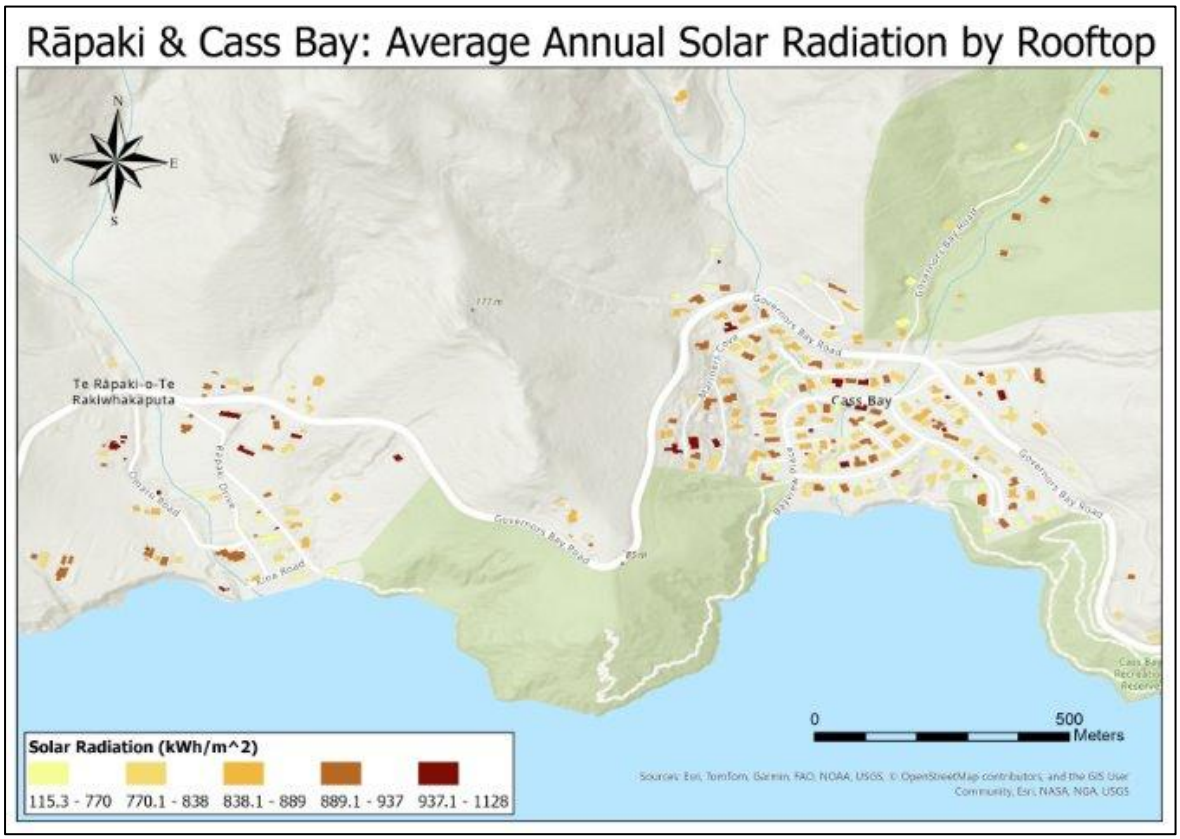


Figure 8: This shows a map of average annual solar radiation by rooftop in Rāpaki and Cass Bay.

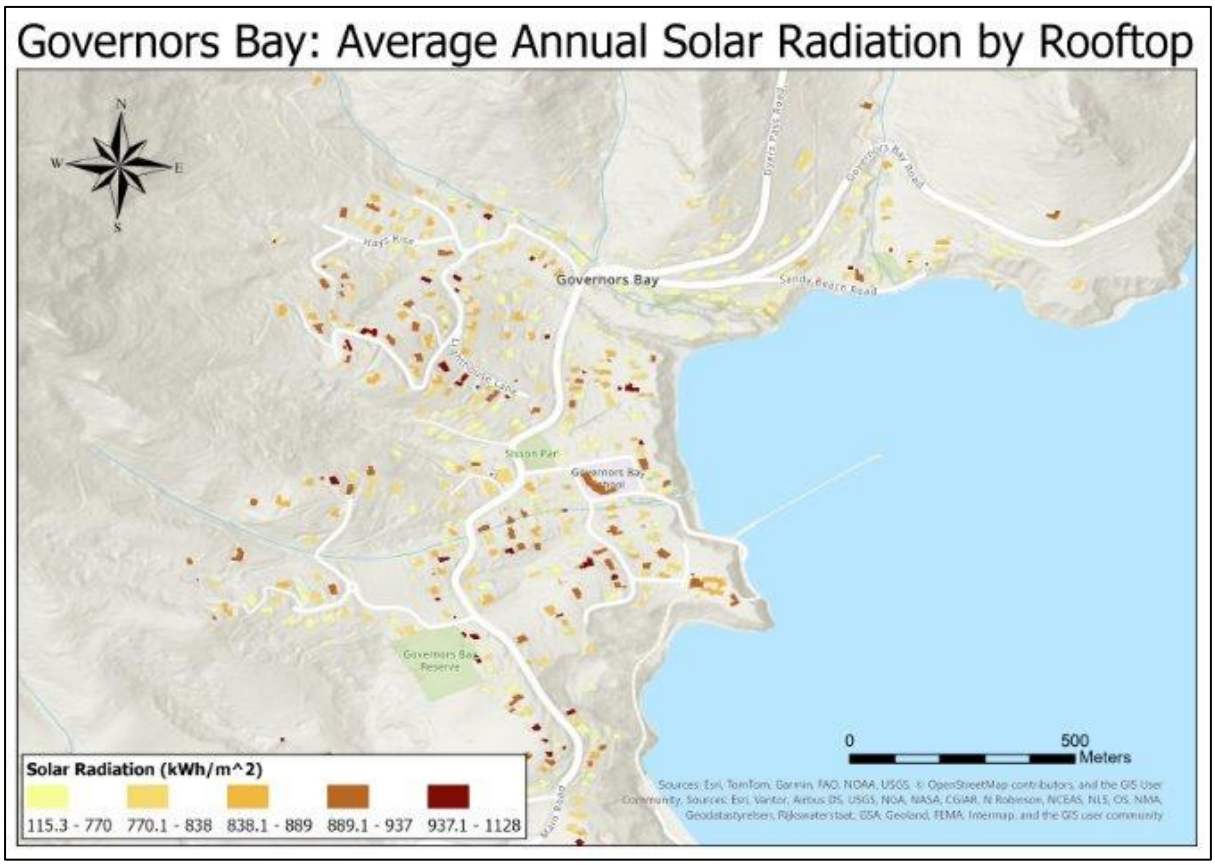


Figure 9: This shows a map of average annual solar radiation by rooftop in Governor's Bay.

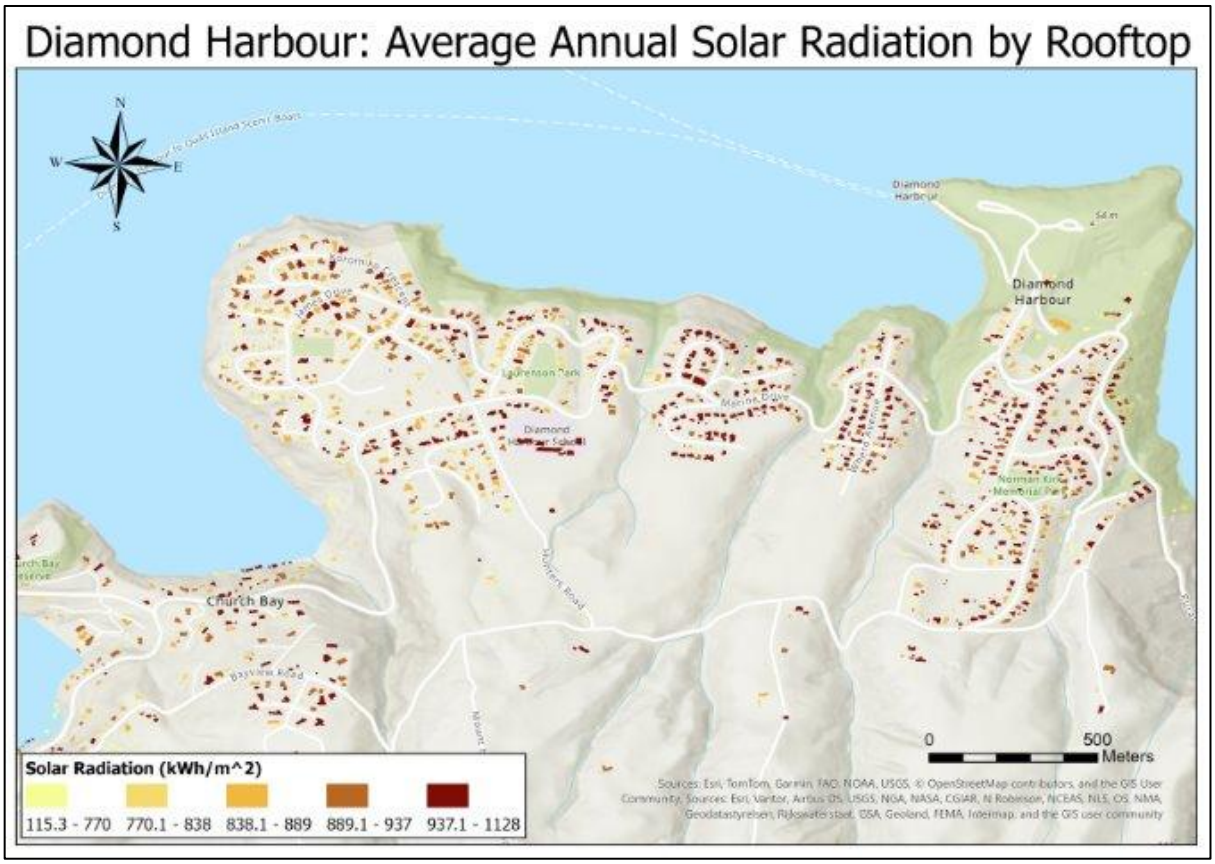


Figure 10: This shows a map of average annual solar radiation by rooftop in Diamond Harbour.

Figure 7 shows on average, the north-east side of Lyttelton received more solar radiation annually than the west side. Hawkhurst Road, found on the west side of Lyttelton has low radiation, whereas Foster Terrace, on the east side of Lyttelton, has high radiation. Figure 3 shows Foster Terrace has many existing solar installations while Hawkhurst Road has none. Solar panel installations are clustered on the north-east side of the suburb, reflecting the high solar radiation distribution present.

Figure 8 shows both Rāpaki and Cass Bay have high solar potential, with many darker coloured buildings. Table A2 reinforces this, showing Cass Bay has the second highest filtered mean with a value of 1054.21 kWh/m^2. Figure 4 shows the existing solar panels in these areas are minimal, with only one solar installation in Rāpaki and nine in Cass Bay. These are areas that could further develop.

Figure 9 shows Governors Bay has a main area of low potential on the west end of Sandy Beach Road. This is due to the steep rise in topography directly north, shown by the darker grey hill shading. The rest of the suburb features high potential with many darker coloured buildings. Figure 5 shows solar panel installations appear dispersed in Governor's bay, which is reflective of solar radiation distribution in Figure 9.

Figure 10 shows Diamond Harbour stands out as the area with the highest average solar radiation. Majority of the suburb features darker colours, with houses in the top 20% of solar radiation values. Unlike Lyttelton, the surrounding mountains lie to the south, meaning they don't obstruct the sunlight from reaching Diamond Harbour. Differences in spatial distribution of solar radiation show the influence of the surrounding topography on the amount of sunlight a suburb receives as exhibited in Figures 7-10.

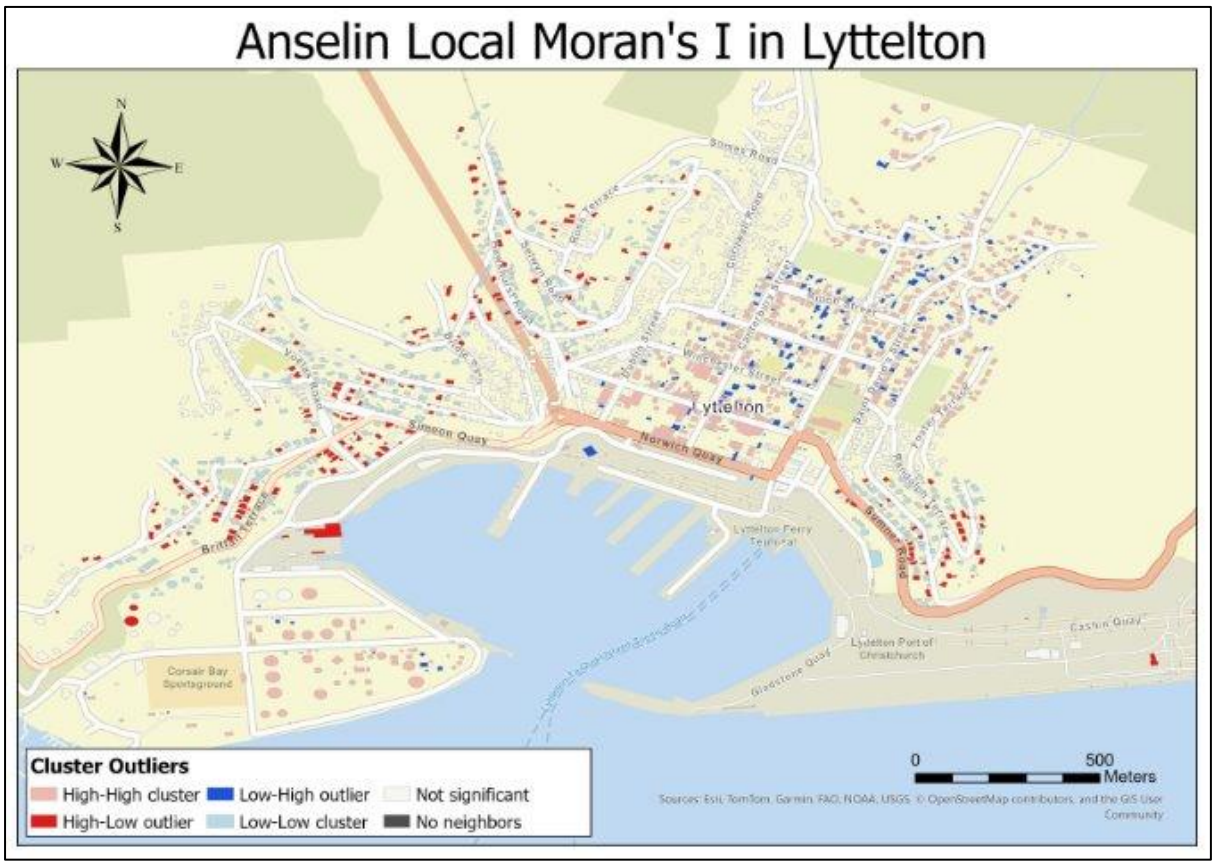


Figure 11: This map shows an Anselin Local Moran's I run in Lyttelton, using the raster solar radiation layer displayed in Figure 7. Each coloured building represents a statistically significant average solar radiation value to a p-value of 0.05.

Figure 11 shows Lyttelton has a significant cluster of low radiation values to the west side and a significant cluster of high radiation values on the north-east side of the harbour. Amongst the Low-Low cluster, bright red buildings represent a significant high radiation value surrounded by significantly low radiation values, and the opposite for bright blue buildings. Bright red outliers could be a result of the building being taller, therefore capturing more radiation than neighbouring buildings. The bright blue outliers could be a result of shading from nearby trees or buildings, decreasing the radiation the building receives. Figure 11 demonstrates although there is spatial pattern of shade and sun in Lyttelton, houses on the shady side still receive high amounts of solar radiation and may be suitable for installation.

Table A2 highlights around 65% of rooftop area is theoretically usable for solar panels in Whakaraupō considering roofs with a slope less than 60 degrees, non-south roof orientation, and roof area of at least 14m^2. Cass Bay has the highest percentage of filtered over theoretical sum, meaning it retained the most pixels after the radiation raster was filtered for slope, aspect, and area. Perhaps this is due to Cass Bay being a more modern area, with houses built considering rooftop solar suitability.

Lyttelton has the highest solar radiation sum, with a value of 94.32GWh, however, Diamond Harbour has the highest filtered mean, with a value of 1054.21kWh/m^2. Lyttelton is a larger

suburb with more rooftops, so the filtered sum is larger than Diamond Harbour. The filtered mean reflects the average solar radiation of each pixel within each suburb. Diamond Harbour has less buildings, but on average roofs receive more sunlight than Lyttelton so the filtered mean is higher. Table A2, paired with Figures 7-10, show that all five suburbs would benefit greatly from increase solar panel installation and have the potential to produce large amounts of solar power.

2.3 Limitations

Creating existing solar installation and physical solar potential maps involved a few limitations: atmospheric conditions, building outline offset, imagery dates and differentiation of solar panels from skylights on rooftops. Atmospheric conditions affected both imagery and the potential modelling. The solar potential model assumed a clear sky average, which could be over representing solar potential. Building outlines were slightly offset due to the angle the aerial imagery was taken, with taller buildings having more offset, known as radial displacement. Another major limitation was manual detection of existing solar panels, where panels could be mistaken for skylights. This introduced potential for misclassification and errors in the estimated number of existing solar panels.

Chapter 3 – Socio-economic influences on household solar uptake

3.1 Literature Review

Rooftop solar adoption is influenced socioeconomic, demographic, and built-environment characteristics alongside physical solar resource. Tao et al. (2024) examined rooftop PV adoption at the census-block level in Nagoya, Japan, using variables including technical solar potential, building density, detached-house ratio, homeownership, and age structure. They found positive associations with technical potential, building density, and detached (standalone) housing, while older age groups were negatively associated with adoption and home ownership was not statistically significant. Literature reviewed by Tao et al. also reports inconsistent relationships for factors such as income, homeownership, and population density across different locations. These findings informed the selection of demographic and built-environment variables examined in Whakaraupō, which were tested against observed PV uptake at the Statistical Area 1 (SA1) level.

Physical suitability alone also does not determine household adoption. Dillman-Hasso and Sintov (2025) found that homeownership, costs, and expected benefits influenced rooftop solar adoption, while renters and lower-income households may face barriers even where solar is technically available. This supported considering socioeconomic characteristics alongside physical solar potential and observed uptake.

3.2 Methods

Demographic characteristics were analysed using 2023 Census data at the SA1 level across residential areas of Lyttelton, Governors Bay, and Diamond Harbour (n = 38, Table A3). Non-residential areas or unsuitable SA1, including the Lyttelton port and harbour area and areas lacking sufficient rooftop imagery, were excluded as seen in Table A4. Census counts were converted into rates using appropriate population or dwelling denominators as shown in Table 2.

Table 1: Variables tested in Exploratory regression.

Variable	Construction
Photovoltaic (PV) uptake (%)	PV-equipped residential buildings / occupied private dwellings × 100
Median household income	2023 Census median household income
Homeownership (%)	Owned/partly owned + family trust / stated tenure × 100
Detached housing (%)	Separate houses / occupied private dwellings × 100
Population 60+ (%)	Residents aged ≥60 / population with stated age × 100
Degree-qualified (%)	Bachelor's degree or higher / population with stated qualification × 100
Residential duration	Mean years at usual residence
Building density	Building footprints / SA1 land area (km²)
Solar potential	Mean building-level Global Average within each SA1
Average household size	Mean number of usual residents per household
Socioeconomic deprivation	NZDep2023 continuous deprivation score

Mean rooftop solar potential was calculated using building-level Global Average solar-radiation values within each SA1. Global Average was preferred to Global Total to avoid conflating solar resource with building size.

ArcGIS Pro *Exploratory Regression* was used to assess combinations of candidate explanatory variables. Models were compared using adjusted R^2, AICc, coefficient significance, VIFs, residual normality, and residual spatial autocorrelation. The final OLS model was selected based on model fit, diagnostic performance, and interpretability.

3.2 Results

Exploratory regression identified a preferred model containing homeownership rate, detached-house rate and the proportion of residents aged 60 years and over. The model had an adjusted R^2 of 0.231 and an AICc of 227.54. Models incorporating socioeconomic deprivation produced slightly higher adjusted R^2 values but poorer AICc and the deprivation coefficient was not statistically significant.

Table 2: Final OLS model of residential PV uptake.

Variable	Coefficient (β)	p-value	VIF
Homeownership rate	−0.165	0.114	2.34
Detached-house rate	**+0.414**	**0.004**	1.93
Population aged 60+	**+0.193**	**0.037**	1.30

n = 38; R^2 = 0.294; adjusted R^2 = 0.231; AICc = 227.54; overall model p = 0.0027

The final model explained approximately 29% of variation in residential PV uptake and was statistically significant overall (p=0.0027). Detached-house rate showed the strongest positive association with uptake (β = 0.414, p=0.004), while the proportion of residents aged 60+ was also positively associated (β = 0.193, p=0.037). Homeownership had a negative but non-significant association (β = -0.165, p-0.114).

Model diagnostics indicated no major violations of OLS assumptions. VIFs were below 2.4, residuals showed no significant departure from normality, and residual spatial autocorrelation was not significant (Moran's *I* = 0.064, p = 0.754).

3.3 Regression Limitations

The regression was limited by the small samples of 38 SA1s and the use of aggregated Census data, meaning area-level associations should not be interpreted as individual household behaviour. There was a temporal mismatch between 2023 Census data and later 2025 PV observations, while likely influences including household finances, environmental attitudes, electricity demand, roof condition, and installation timing were unavailable. Manual PV identification and the use of PV-equipped buildings relative to occupied private dwellings may have measurement error.

Chapter 4: Community Energy; Constraints, Resilience & Equity

4.1 Literature Review

Community energy arrangements, such as virtual power plants (networks of home solar-and-battery systems managed together), can keep communities powered when the main grid fails. During Cyclone Gabrielle, around 3,000 SolarZero customers kept their power on while about 225,000 households lost theirs, and some homes became informal hubs helping neighbours (Pyle, 2025). Community batteries similarly stabilise the local grid and store surplus solar for later use (Climate Connect Aotearoa, 2025). However, the Electricity Industry Participation Code only allows one-to-one customer to retailer contracts, blocking peer-to-peer sharing, and high upfront costs put solar out of reach for renters and low-income households (Climate Connect Aotearoa, 2024, 2025).

4.2 Methods

For each case, how the source reported its arrangement was recorded: how it worked around or within legislation, how energy was traded or shared, and which challenges arose and which did not. Each case was then assessed against resilience and equity parameters. The entire analysis has been displayed in Tables A5 – A9 as appendices, however, the most applicable of these parameters for Whakaraupō within our understanding of LETS aspirations are discussed below.

As community led solar is new in New Zealand, there was a shortage in diverse research literature. Therefore, some sources used were grey literature, which have much larger room for error in bias.

4.3 Results

The cases show that community solar in New Zealand is possible, but the legislation limitations have more influence than technical feasibility.

Most arrangements reported either staying within the one-retailer rule or needing an exemption from it. Standard export is the simplest option because it stays within the rule, but it pays on average 8c-16c per unit exported and charges about 30c-38c to buy (Wallis, 2026), this also applies overseas. Schemes that move energy between separate homes, like the Kāinga Ora and Ara Ake pilot in Wellington and a North Island gifting pilot, ran under an exemption. That exemption is currently due to expire in 2028 (Van Der Merwe, 2025).

How energy moved depended on what each scheme was trying to do. According to Wade et al. (2025) separate sharing (energy given away), trading (energy swapped for something, like a basket of vegetables) and selling (energy sold for money). Much of what is marketed as sharing, is selling. In NZ, sharing and selling have both been done, but trading needs peer-to-

peer platforms that do not exist yet. Community solar farms avoid this by selling to the market and crediting members' accounts (Barrett et al., 2025).

Cost and timing came up most in terms of challenges. Batteries are expensive and in the cases of community energy a subsidy was usually needed, and network charges can wipe out the savings unless there is a local-use tariff (Climate Connect Aotearoa, 2024, 2025). Solar peaks around 3pm where many people aren't home, so some form of storage is needed. Cohaus avoided an estimated $150,000 transformer upgrade by using load control and timers for their hot water cylinders instead of a battery (Livesey et al., 2025). Pooling batteries across buildings was modelled to cut shortfall hours by 97%; shortfall hours are the hours each year when a community can't meet its own power needs from the energy it generates and stores, so it must rely on the grid or go without (Apperley & Toki, 2025). Priority access for vulnerable households did not reduce performance, although this was only modelled (Wade et al., 2025).

During Cyclone Gabrielle, homes with solar and batteries stayed powered, and some supplied their neighbours (Pyle, 2025a). A hub site that can carry extra load, like the marae modelled at Motairehe, is another option (Apperley & Toki, 2025). Equity worked best when it was built in from the start. Kāinga Ora put solar on tenants' homes without batteries, so the money reached more households, and export income paid for further installs (Van Der Merwe, 2025). These community-owned assets included people without suitable roofs. The risks are schemes closing into "friend groups", and people joining only because they cannot afford energy from the normal market if increased widespread solar use, and less "legacy-grid" demand causes grid prices to spike ridiculously high (Wade et al., 2025).

Chapter 5: Restrictions of the Distribution Network

5.1 Methods:

Access to electricity network data was a major limitation of the project. Relevant data was requested through a LGOIMA process, which when provided required additional interpretation and processing. Restrictions around sharing of this data resulted with inability to directly assess local grid capacity as a constraint on future rooftop solar uptake. As a result, this study relied on openly available data and case studies to inform LETS of network restrictions.

5.2 Results:

Physical rooftop solar potential does not reflect how much generation the electrical network can accommodate. Hosting capacity represents the total quantity of distributed generation that can be installed without overwhelming the networks operating limits. Increasing photovoltaic (PV) generation can introduce reverse power flows, where low voltage networks were traditionally designed for electrical supply straight to consumers from centralised generation. Therefore, voltage rise, alongside conductor and transformer overloading are key constraints to increased PV installation (Watson et al., 2016; Rajabi et al., 2022).

Hosting capacity varies spatially, being driven by feeder length, conductor impedance, transformer capacity, customer demand and the location and concentration of PV instalments (Crownshaw et al., 2016; Koirala et al., 2022). NZ research proves that hosting capacities are driven by network types and network conditions, and that one capacity derived from literature cannot represent our local limits (Watson et al., 2016).

These findings suggest that Whakaraupō's solar potential should be interpreted with respect to local network conditions. Areas with high solar potential may face constraints on installation of additional generation. However, to solve this issue, hosting capacity can be increased through voltage management, battery storage, smart inverters, changes in electricity usage and targeted infrastructural upgrades (Petinrin & Shaaban, 2016; Rajabi et al., 2022).

Chapter 6: Discussion

Lyttelton has 1,518 dwellings (Stats NZ, 2023). A solar project for the community could become a reality if significant funding is acquired. The main limitation of this research was that data on the local transformer locations and the houses connected to each one couldn't be obtained (see Chapter 5). Without them, a community microgrid area cannot be effectively defined, so emphasises on a community hub was applied making it easier to achieve in terms of legislation, technical feasibility and cost. This section identifies options for solar system installations, drawing on the findings of each chapter.

6.1 Potential Community microgrids in Whakaraupō

Blocks with high-potential roofs (Figure 7) and existing panels (Figure 3) are the natural place to start. The regression results support this: uptake was most strongly linked to the proportion of standalone houses, and high solar potential did not necessarily mean high uptake. Roofs with high potential but few space for panels are therefore worth prioritising alongside those that already have solar. Selwyn Parade and Ross Terrace (Figure 12, point A), the Somes Road cluster (Figure 12, point B) and possibly anywhere along the east side around Foster Terrace (Figure 12, point C) are suitable candidates for early microgrids. Transformer data would allow a more specific choice of homes.

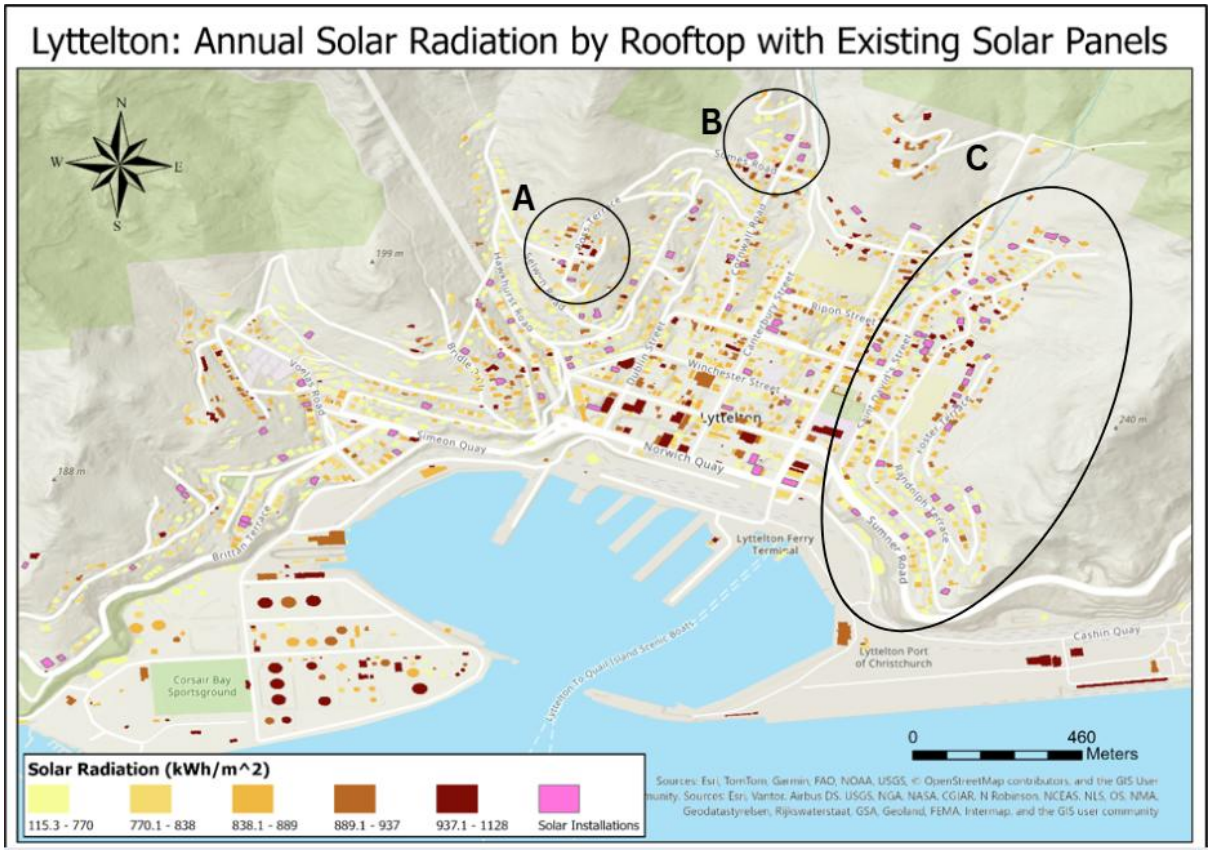


Figure 12: This shows the average solar radiation map from Figure 7, combined with existing solar panels from Figure 3. Marked areas are ideal spots for the first community microgrid in Lyttelton harbour.

Households that already have solar should be encouraged rather than pressured and given a fair way to join. If other households receive free panels to share energy, there should be a pay-back schedule for those without panels, or compensation for those who paid for their own years ago. Households without high-potential roofs also need to be included, including those in attached or constrained dwellings who may face barriers whether they are willing or able to pay. Shared solar could help here. The regression data are aggregated, so they cannot show household finances or attitudes. Consultation (advertising, education and a survey) is therefore essential, and an information night followed by anonymous feedback would be a useful future study. One arrangement could include up to 20 dwellings (Livesly et al. 2026). How surplus is shared is a governance choice rather than a hardware one. Wade et al. (2025) found that mutual-support approaches performed better than profit-first ones, and that giving vulnerable households' free priority access did not reduce performance (modelled, not empirical).

6.2 Potential Community hubs in Whakaraupō

The Community Centre, Community Boardroom and Recreation Centre all have moderate to high solar potential. They also appear straightforward under the current legislation because they would not supply power to other dwellings. One issue is that grid-tied systems must switch off automatically during an outage, to protect people repairing the lines from power flowing back in (DaD Electrical, 2024). A solar system with a battery can instead run independently. This would let the hubs stay open as places where people can charge phones or keep warm in an outage, and they could also be a source of investment.

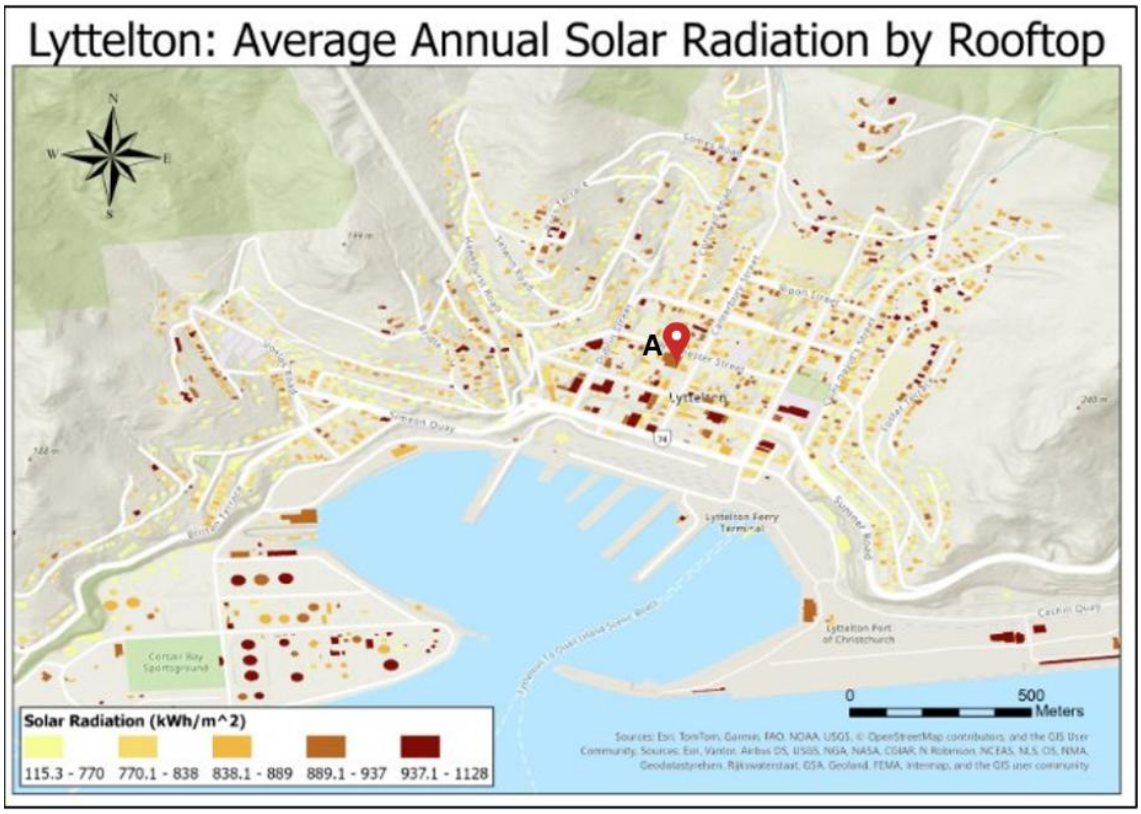


Figure 13: This shows the average solar radiation map from Figure 7, with a location marker (point A) showing where the community "safehouse" buildings are located on the solar radiation map.

6.3 Gebbie's Pass grid-connection

A former wind farm site at Gebbie's Pass has a 500-kW connection to Orion's lines but no turbine (Sam Kivi, personal communication, 16th September 2026). With the landowner's written approval, solar panels could be installed there. The power could be a council-funded asset, a shared asset between LETS and the landowner, or sold back to the grid by LETS to fund more solar installations.

Recommendations

If LETS wishes to extend the demographic analysis undertaken in this study, recommendations include collecting household-level data through a targeted community survey which was beyond the scope and timeframe of our project. This would allow future analysis to move beyond aggregated SA1 characteristics and examine factors more directly related to household solar adoption, such as financial capacity, environmental attitudes, electricity use, housing circumstances and willingness to invest. This would help LETS better target future community energy initiatives and identify where alternative models, such as shared or community solar, may be most useful.

For future research, obtaining detailed distribution network data from Orion to estimate the capacity of additional rooftop solar across Whakaraupō would hold value. Information including transformer ratings, feeder characteristics and electricity demand would support new modelling of suggested increased solar uptake. This would allow for testing of different demand scenarios helping estimate the local hosting capacity as well as identifying areas/infrastructure that bring constraint and determining where network upgrades may be needed.

Conclusion

This research investigated the potential for rooftop solar generation across Whakaraupō, focusing on Lyttelton, Rāpaki, Governors Bay and Diamond Harbour. Our main findings are that physical potential for solar is substantial, but only partially being used. Solar potential varies spatially due to complex terrain of Banks Peninsula while household uptake is shaped by socioeconomic factors. Community scale energy sharing provides a way around this; however, it is limited by governance.

Solar installations while present (chapter 1), only capture some of the available PV potential identified, leaving room for growth. Rooftop potential (chapter 2) is not evenly spread at each location. Rooftops on the north-east side of Lyttelton receive substantially more incoming solar radiation than the western side where topography and shading reduce output. Rāpaki, Cass Bay, Governors Bay and Diamond harbour, all show varying patterns of solar potential, but consistently exhibit exposed areas with high potential and valleys with reduced potential.

This study exposed that physical potential is not the only constraint on installations. Socioeconomic factors of detached housing and age have influence on current installation patterns, while hosting capacity also constrains. Sharing energy could help overcome these barriers by letting households with unsuitable or unaffordable rooftops benefit from solar generated elsewhere, however current legislations limit this option.

Together, these findings highlight that solar potential in Whakaraupō is high, but there are socioeconomic and governance barriers along with physical barriers restricting maximisation. For LETS, this suggests focusing on higher potential areas and exploring community-based models that could benefit lower potential areas within the limits of the current legislation.

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Appendix

Table A1: Table showing the number of houses with solar installations/number of houses to give a % of houses in each area with existing solar panels

Location	Number of solar installations/Number of houses	% of Houses with Solar Panels
Lyttelton	80/1518	5.27%
Rāpaki	1/39	2.56%
Cass Bay	9/118	7.62%
Governors Bay	31/393	7.89%
Diamond Harbour	62/1053	5.89%

Table A2: This table shows values corresponding to the theoretical and filtered theoretical solar potential of Whakaraupō. These values were calculated using the Raster Solar Radiation Tool on ArcGIS Pro. The Sum columns represent the total solar radiation hours over an annual period, and the Mean columns represent the Sum divided by the total count of rooftop pixels (or rooftop area).

Suburb	Theoretical Sum (GWh)	Filtered Sum (GWh)	Filtered / Theoretical * 100	Theoretical Mean (kWh/m^2)	Filtered Mean (kWh/m^2)
Lyttelton	144.85	94.32	65.12%	911.02	995.66
Diamond Harbour	79.17	52.36	66.14%	965.86	1054.21
Governors Bay	33.07	21.56	65.20%	873.59	971.89
Cass Bay	17.60	12.01	68.24%	934.88	1013.80
Te Rāpaki-o-Te Rakiwhakaputa	9.53	5.93	62.23%	885.28	986.50

Table A3: SA1s included in the residential solar analysis

Area	Included SA1 codes
Governors Bay	7026600, 7026602, 7026603, 7026604, 7036135, 7036136
Lyttelton	7026606, 7026607, 7026608, 7026609, 7026610, 7026611, 7026613, 7026614, 7026615, 7026616, 7026617, 7026618, 7026623, 7026624, 7026625, 7026626, 7026627, 7026628, 7026629
Diamond Harbour	7026619, 7026630, 7026631, 7026632, 7026633, 7026634, 7026635, 7026636, 7026637, 7026638, 7036137, 7036138, 7036139

Table A4: SA1s excluded from the residential solar analysis

Area	SA1 code	Reason for exclusion
Lyttelton	7036145	Primarily comprises the Lyttelton port and harbour area and contains no meaningful conventional residential dwelling stock. Although nine usual residents were recorded in the 2023 Census, its land use and geography made it unsuitable for comparison of household rooftop solar uptake.
Diamond Harbour / Port Levy	7026639	Excluded because the SA1 covered a very large area that was not considered comparable with the more compact residential SA1s and could distort spatial analysis. It was also not identified in the Census lookup table used for the demographic analysis.
Diamond Harbour	7026640	Excluded because no usable Census data were available for the analysis.
Teddington	7026605	Excluded because suitable rooftop imagery was not available to map existing residential PV installations consistently.
Teddington	7026612	Excluded because suitable rooftop imagery was not available to map existing residential PV installations consistently.

Community microgrids

Table A5: Community microgrids; A small cluster of houses forming their own grid behind the meter, with or without a shared battery.

Governance / Methodology	Description	Technical Feasibility / needs	NZ trialled?	References
Off-grid, distributed batteries	Each building keeps its own solar and battery, but generation and storage across the group are pooled and allocated together rather than run as separate systems.	Needs hourly load modelling per building and a coordination system to direct surplus between units; pooling cut shortfall hours by 97% versus running the same total resources separately.	Yes	Apperley & Toki (2025), Ch. 10
Grid-connected, single shared central battery	Multiple households connect through a microgrid built around one large communal battery, instead of each household owning a smaller one.	Needs dedicated transmission and system-management technology, a compliance process, an ownership structure, and a decision on single vs	Not at scale; feasibility and technical modelling stage only.	Mohseni & Brent (2025); Aperahama, Penny & Taptiklis (2025), Ch. 11

		multiple grid connection points.		
Grid-connected, no battery; load control + thermal storage instead	One shared connection point to the network; instead of a battery, demand is capped with a load controller (non-essential loads shed first), and surplus power is stored as heat in hot water cylinders timed to cover the morning peak.	Needs a load-control system and per-unit metering/billing; this substitution avoided an estimated $150,000 transformer upgrade.	Yes, currently operating (Cohaus, Auckland).	Livesey, Welch & Yeats (2025), Ch. 5

Community hub

Table A6: Community Hubs; One larger, centralised solar or storage asset used for community-wide resilience.

Governance / Methodology	Description	Technical Feasibility / needs	NZ trialled?	References
Standalone community hub with battery	A single, community-owned battery separate from individual rooftops stores local surplus centrally and discharges it back to members, rather than each household storing its own. Saves surplus for homes without batteries.	Needs smart energy management and bidirectional flow; typically needs subsidy, as batteries are expensive; distributor tariffs on energy in/out can erode savings unless a local-use tariff exists.	Yes; NZ's first community battery, running at roughly 30% below market rates.	Climate Connect Aotearoa (2024, 2025); Evans et al. (2024)
Community-scale solar farm, centrally managed returns	One larger generation asset is owned and managed by a community; its output is sold to the spot market, and proceeds are credited straight to members' power accounts, or as revenue to LETS.	Needs an operator to manage market sales and distribute returns; removes install-cost and roof-suitability barriers for members.	Yes.	Barrett, Barrett & Te Kanawa (2025), Ch. 9
Aggregated home batteries acting as a virtual hub (VPP)	Instead of one large asset, many household batteries are aggregated and centrally directed as if they were one, charged to maximum ahead of	Needs a VPP control platform and an operator; ~10.8 kW per home identified as a workable	Yes; real event, Cyclone Gabrielle, Feb 2023.	Pyle (2025a), Ch. 12

	forecast severe weather so individual homes can extend supply to neighbours during an outage.	sizing benchmark for winter resilience.		
Hub embedded within a microgrid	One site within a wider microgrid; in one specific case, a marae is designated to absorb irregular high loads and hold backup capacity for the rest of the group, rather than spreading that role evenly across every building.	Needs separate load modelling for the hub site, since its usage pattern differs from an ordinary household.	Modelled (Motairehe, Aotea/Great Barrier Island).	Apperley & Toki (2025), Ch. 10

Energy sharing and trading: important distinctions & governance

Table A7: Important flags identified for governance

Governance / Methodology	Description	Technical Feasibility	NZ trialled?	References
Energy "Sharing"	Giving energy for no material return; the recipient only pays the transportation fee. Wade et al. treat this surplus as a shared commons rather than a commodity: the point is building community connection, not revenue.	Needs an exemption to NZ's one-retailer rule (multiple trading relationships) and shared or community storage to time-shift surplus, since peak sun (~3pm) rarely matches when households are home.	Yes, under exemption; Kāinga Ora × Ara Ake pilot (Wellington), and a North Island gifting pilot.	Wade et al. (2025); Climate Connect Aotearoa (2024); Van Der Merwe (2025), Ch. 7
Energy "Trading"	Giving energy for a material return, e.g. vegetables for 2kWh; still relationship-based, but a return is expected. Wade et al. warn that trading only within existing friendships can turn a scheme into a closed clique, so explicit rules for including newcomers are needed.	Needs peer-to-peer trading technology and mini-market platforms that don't yet exist in NZ, plus the same multiple-trading-relationship exemption.	Not directly; named as a desired outcome by the Electricity Authority and MBIE, but no NZ “mini-market” exists yet.	Wade et al. (2025); Mohseni & Brent (2025); Aperahama, Penny & Taptiklis (2025), Ch. 11
Energy "Selling" (often marketed as sharing)	Giving energy for monetary gain. Wade et al.'s key point: a lot of what's marketed as "energy sharing", like renting out solar	Works within the existing one-retailer relationship, making it the most straightforward option and	Yes; there are standard feed-in arrangements (roughly 8c/kWh	Wade et al. (2025); Climate Connect Aotearoa (2024)

	capacity, is really selling, since it's profit-driven rather than connection-building.	technically part of why it dominates. However, whether the grid can handle every home selling energy back is a whole other story.	sell vs 35c/kWh buy).	
Profit-first vs solidarity-first allocation	The rule a community uses to decide where surplus goes. Wade et al. found systems optimised purely for profit perform worse overall, while those prioritising mutual support benefit everyone more. And giving vulnerable households free priority access didn't hurt performance.	A governance/software setting, not so much a hardware requirement, the same physical system can run either rule.	Not tested, Wade et al. is a modelled (agent-based) study, not empirical.	Wade et al. (2025)
Policy-driven (coerced) participation; a risk, not a scheme	Wade et al.'s warning: if governments cut the feed-in tariff (what you're paid for exporting), as desired by some sole energy generators, this will cause gentailers to sell energy at a more expensive rate, local sharing schemes become the cheaper option by default. The risk is that people end up joining not because they value community energy, but because they've been priced out of the standard market; which undermines the legitimacy of calling it "community" energy and tends to pull in the people least able to absorb switching or admin costs, working against equity rather than for it.	A design risk to weigh when advocating for tariff or pricing changes.	Not documented as happening yet, but NZ's existing ~8c export / 35c import gap already points this direction.	Wade et al. (2025); Climate Connect Aotearoa (2024)

Resilience parameters

Table A83: Parameters identified for creating energy - resilience for a community.

Parameter	Description	NZ trialled?	References
Supply shortfall hours	How many hours a year the community can't meet its own power needs. Pooling generation and storage across the group; instead of each household relying only on its own.	Modelled for am NZ community.	Apperley & Toki (2025), Ch. 10
Keeping the power on during a grid outage	Whether a household or community can keep supplying itself when the main power grid fails. Homes with solar-plus-battery stayed powered, and some supplied neighbours, during the 2023 outage that cut off 225,000 households.	Yes, Cyclone Gabrielle.	Pyle (2025a), Ch. 12
Enough battery to last through a real outage	Whether stored energy is actually enough to see a household through a realistic power cut. Around 10.8 kW was found to be enough to support a home through winter.	Yes.	Pyle (2025a), Ch. 12
Charging up before bad weather hits	Whether batteries are deliberately topped up ahead of a forecast storm, and kept charging once it passes, rather than sitting at whatever charge level they happen to be set at at all times.	Yes.	Pyle (2025a), Ch. 12
Using more than one energy source	Spreading generation across different renewable sources (e.g. solar and wind) so the community isn't reliant on one weather conditions; this is relevant since droughts affect hydro and low wind affects turbines.	Feasibility work only.	Mohseni & Brent (2025); Aperahama, Penny & Taptiklis (2025), Ch. 11
Less energy lost in transport	Local systems lose less power moving it around than electricity travelling long distances across the national grid. Shorter distances ensure more of what's generated actually reaches the user.	Feasibility work only.	Mohseni & Brent (2025)
Protection from price spikes	Having your own local generation and storage buffers a household from sudden jumps in the wholesale price of electricity, a financial form of resilience, not a physical one.	Feasibility work only.	Mohseni & Brent (2025)
Physically storm-proofing the infrastructure	Strengthening existing power lines and equipment, along with PV equipment, so they survive more severe weather, relevant given NZ's rugged terrain, dispersed population, and increasingly severe storms.	Recommended, not yet widely done.	Pyle (2025a), Ch. 12
Storing surplus to use later, and helping the wider grid	Whether excess power generated at one time of day can be saved and used later, which also takes pressure off the wider network, important because NZ has little storage overall, so the system currently has to generate to match demand in real time.	Yes; community battery trials.	Climate Connect Aotearoa (2024, 2025)

Having a dedicated site that can support others	Whether there's a specific site; a marae or a cluster of homes that can absorb extra load and supply neighbours during an outage, and how many households one such site can realistically cover.	Yes; both SolarZero homes and the Motairehe marae.	Pyle (2025a), Ch. 12; Apperley & Toki (2025), Ch. 10

Equity scheme ideas

Table A94: Parameters identified as increasing equity among a community - technically also increasing resilience.

Governance / Methodology	Description	Technical Feasibility	NZ trialled?	References
Targeted subsidy, solar first for lower-income tenants (no battery)	A landlord (Kāinga Ora) funds solar installs specifically for tenants likely to be priced out otherwise; batteries are deliberately left out so the same budget reaches more households rather than fewer.	Needs a multiple-trading-relationship exemption for standalone homes to buy and sell separately; apartments are simpler, since they already share one connection.	Yes.	Van Der Merwe (2025), Ch. 7; Climate Connect Aotearoa (2024)
Cross-subsidy, export revenue funds further install	Surplus from tenant rooftops is sold back to the landlord, and that revenue funds solar for more homes in energy hardship, turning exports into a self-funding pipeline rather than a one-off grant.	Depends on the same MTR exemption (currently set to expire 2028).	Yes.	Van Der Merwe (2025), Ch. 7
Centralised asset, no rooftop required	A community-owned generation asset credits returns directly to member accounts, so households benefit without owning any equipment, removing install cost, roof suitability and tenancy restrictions in one move.	Needs an operator to manage market sales and account crediting.	Yes.	Barrett, Barrett & Te Kanawa (2025), Ch. 9

Priority allocation for vulnerable households	An allocation rule giving lower-income or less flexible households first access to free or cheap surplus, modelled to not harm overall system performance.	Needs the MTR exemption plus an allocation algorithm in the community's software.	Not formally tested; closest NZ analogue is a gifting pilot.	Wade et al. (2025); Climate Connect Aotearoa (2024)
Explicit membership / inclusion rules	Formal rules for admitting newcomers rather than relying on existing personal networks, prevents schemes closing into cliques and addresses the knowledge gap between newcomers and energy-sector-fluent members.	Administrative/governance, not technical.	Recorded as a lesson learned, not formalised into policy.	Wade et al. (2025); Barrett, Barrett & Te Kanawa (2025), Ch. 9